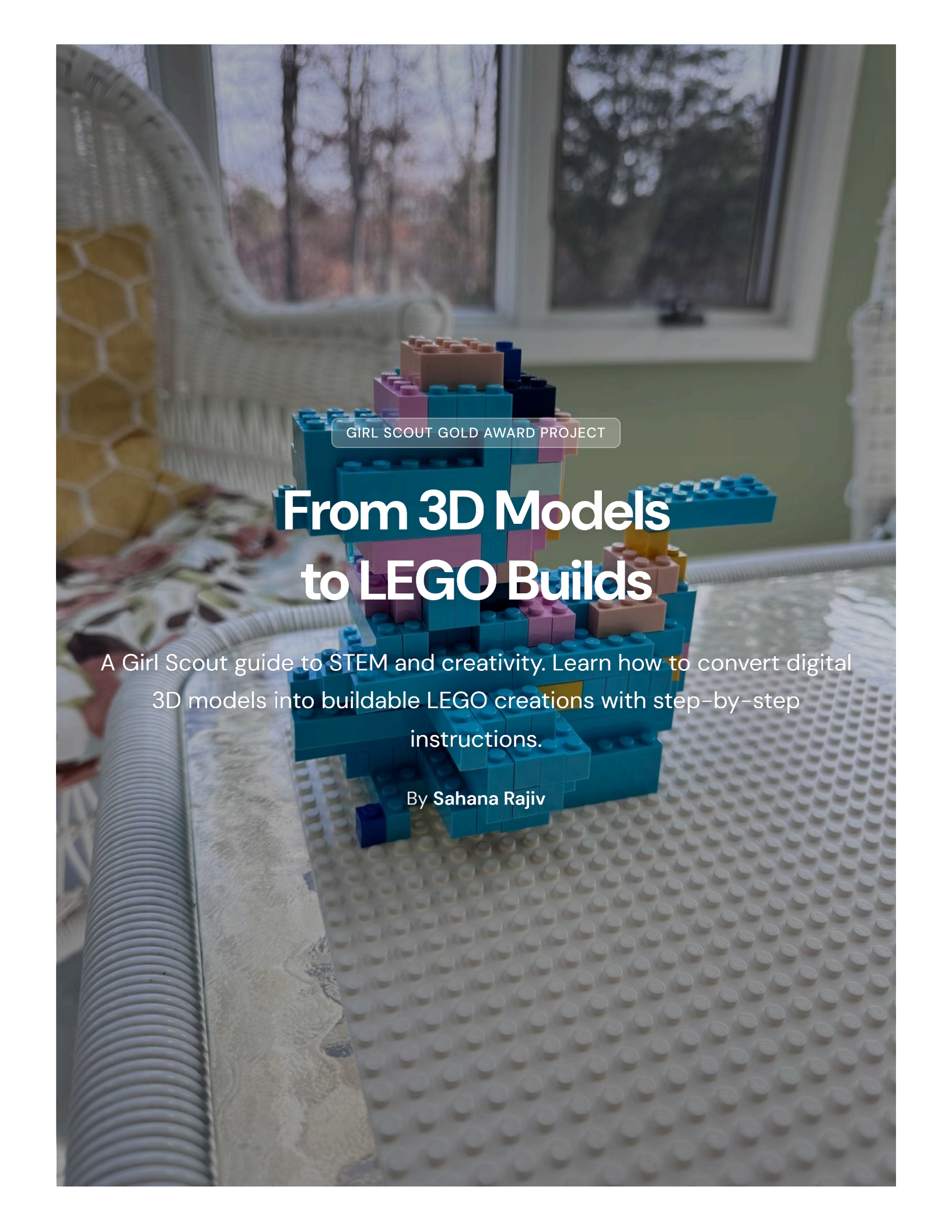




GIRL SCOUT GOLD AWARD PROJECT

# From 3D Models to LEGO Builds

A Girl Scout guide to STEM and creativity. Learn how to convert digital 3D models into buildable LEGO creations with step-by-step instructions.

By Sahana Rajiv

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# Introduction

## Why I Built This Project

For my Girl Scout Gold Award, I wanted to create a STEM project that combined creativity, problem-solving, and real-world impact. I have always loved building with LEGO bricks, but I also enjoy working with technology and coding. I noticed that while there are many amazing 3D models online, it is not easy to turn those digital designs into something people -- especially kids -- can actually build in real life.



### Creativity

Combining digital art with physical building



### Problem-Solving

Tackling complex engineering challenges



### Real-World Impact

Making STEM accessible for younger students



### STEM Education

3D geometry, spatial reasoning, and coding

## What I Created

I created a web-based tool that converts 3D STL files into LEGO models that can be physically built. The program analyzes a 3D model, breaks it into layers, selects LEGO bricks that fit the shape, and generates step-by-step building instructions similar to official LEGO manuals.

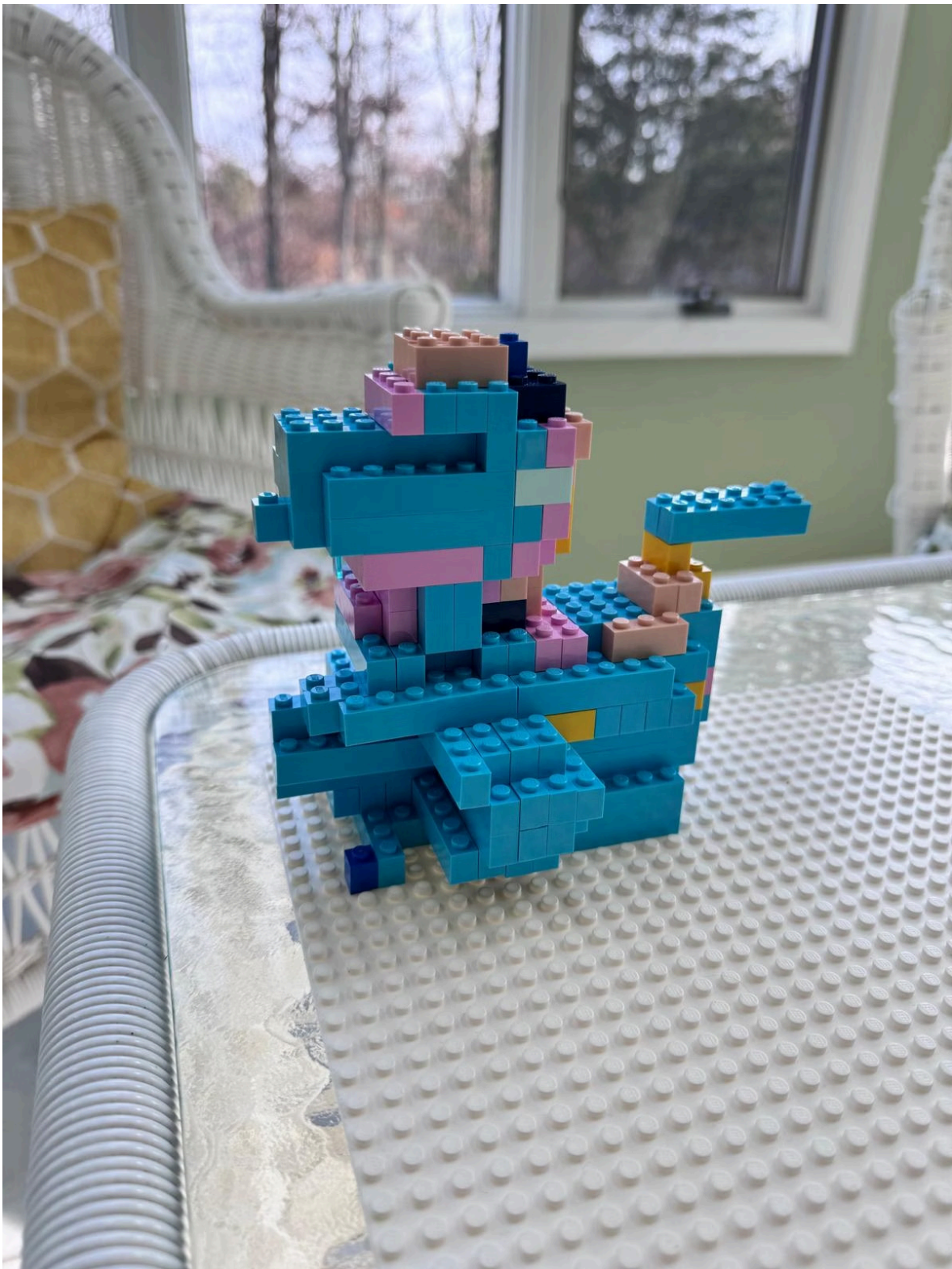
## More Than Just LEGO

This project is not just about LEGO. It is about learning how digital ideas become physical objects. Through this tool, users learn about 3D geometry, spatial reasoning, problem-solving, and computational thinking. My goal was to make advanced STEM concepts accessible, hands-on, and fun -- especially for younger students and Girl Scouts.

## Who This Book Is For

This book explains my project in detail and shows how others can use, modify, and share it. Whether you are a student, a Girl Scout, a troop leader, or a beginner coder, this guide will help you understand how the project works and how to run your own LEGO STEM workshop.





A completed LEGO Rubber Duck created using the STL-to-LEGO converter

## How It Works

### The STL-to-LEGO Converter

For Kids

For Leaders

This chapter explains what the computer does when it turns a 3D picture into LEGO bricks you can build with.

### What Is an STL File?

An STL file is a common type of 3D model file. It describes the surface of a 3D object using tiny triangles. These files are often used for 3D printing, digital design, and modeling software.

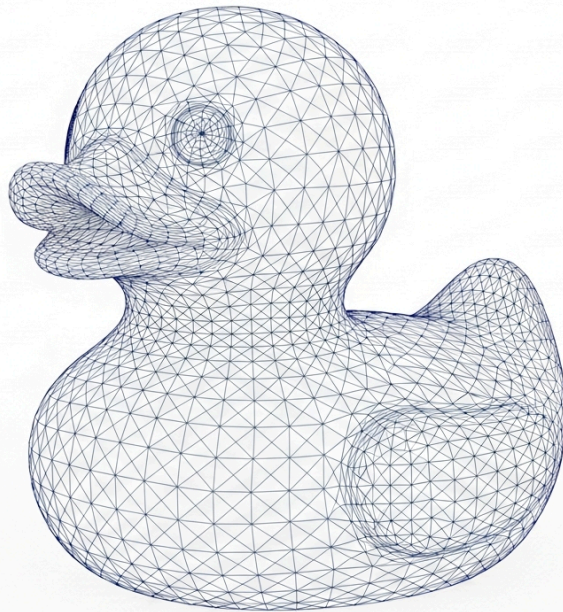


Figure 2.1 -- An STL file describes a 3D shape using thousands of tiny triangles

### Turning 3D Models into LEGO Builds

LEGO models are built from small rectangular bricks, not smooth surfaces. To solve this problem, my program converts the 3D model into a grid of small 3D blocks, similar to pixels but in three dimensions. This process is called **voxelization**.

Each voxel represents a small volume of space that can be filled with LEGO bricks. The program stacks these voxels layer by layer, starting from the bottom and moving upward.



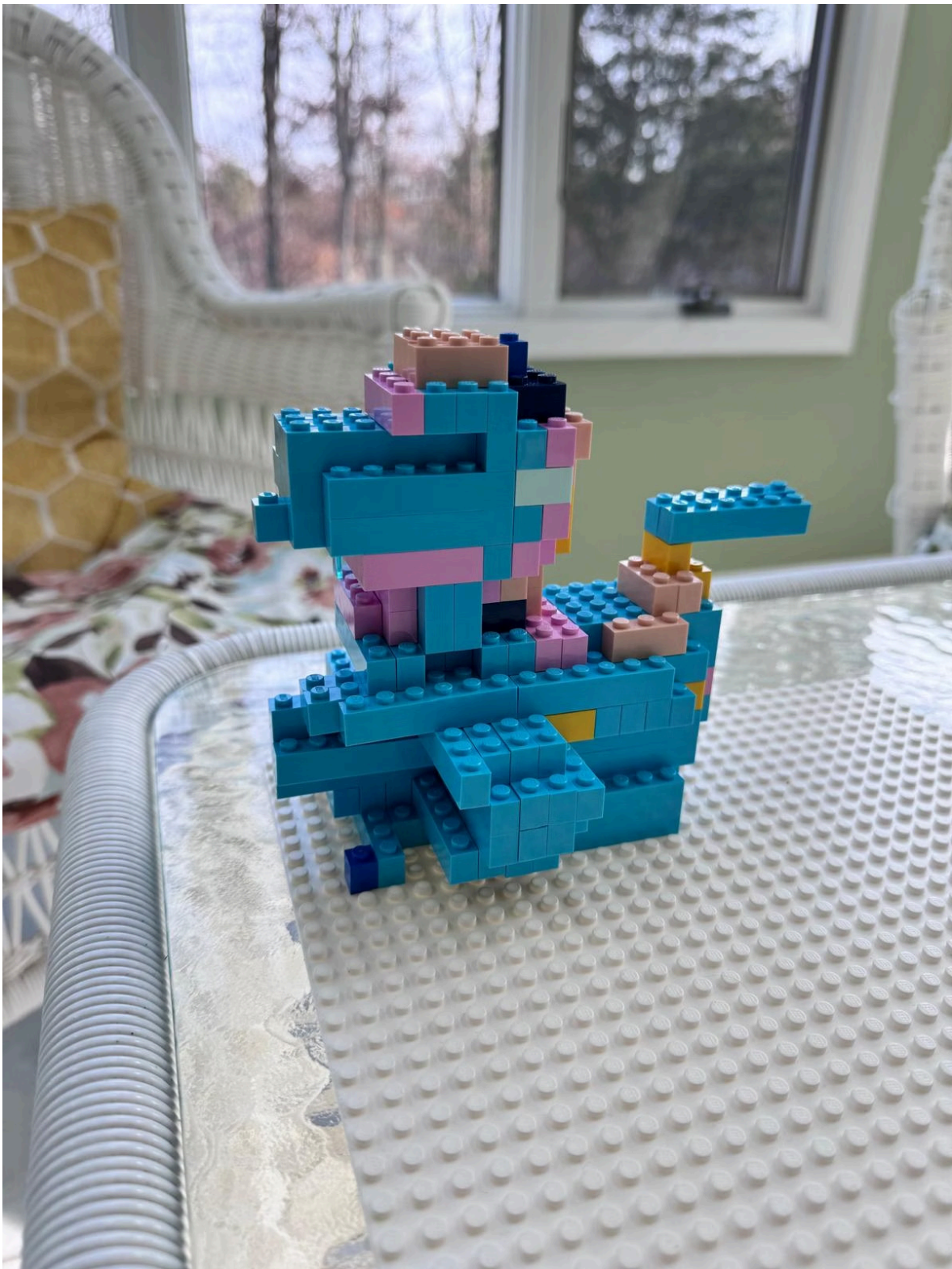


Figure 2.2 -- The finished LEGO duck, built using the converter's instructions

## Key Features of the Converter

The converter does more than just stack bricks:



### Removes Floating Bricks

Every piece is supported so your build won't fall apart



### Smart Brick Selection

Chooses brick sizes that make the build stronger



### Color-Coded Bricks

Assigns colors to help with organization



### Step-by-Step Instructions

Visual guides for each layer of the build



### 3D Preview

Creates a new 3D model of the LEGO version



### Brick Inventory

Complete list of every brick you need

By the end of the process, the user receives a LEGO model that is stable, buildable, and easy to follow.

# Modifying the Code

## A Step-by-Step Guide

This chapter is for anyone who wants to explore or modify the code behind the project.

### 1. Project Setup

Before you can run or modify the converter, you need to set up the project on your computer. Follow these steps carefully.

- Clone the repository from GitHub (or download as ZIP)
- Navigate into the project folder
- Install dependencies with `pip install -r requirements.txt`
- Run the web app with `python app.py`
- Open `http://localhost:5000` in your browser

### 2. Common Modifications

Here are the most useful changes you can make in `lego_1.py`:

- **Change model size** -- adjust the voxel resolution variable (try 8, 12, 16, or 20)
- **Limit brick types** -- remove sizes from the `brick_sizes` list to match bricks you have
- **Change colors** -- update the `color_map` dictionary with new RGB values
- **Toggle stability** -- comment out `remove_floating()` to skip stability checks
- **Customize instructions** -- adjust grid line thickness, fonts, and page layout
- **Change output format** -- modify the pipeline to generate PNG or skip STL export

### 3. Using Cursor AI

Throughout this project, I used **Cursor** -- an AI-powered code editor -- as my coding sidekick.

#### What Is Cursor?

- Free code editor from [cursor.sh](https://cursor.sh)
- Built on VS Code -- looks and feels familiar
- Has a built-in AI chat that can see your entire project
- Ask questions, request changes, or have it write new features
- Like having a coding tutor who can read all your files

#### Setting Up Cursor with the GitHub Project

Here is how to get started using Cursor with the LEGO converter code:



1

### Download and Install Cursor

- Go to [cursor.sh](https://cursor.sh)
- Download the free version (Windows, Mac, or Linux)
- Install it like any other app

2

### Clone the Repository

- Open Cursor, go to File > Open Folder
- Navigate to the Web-App-Lego-Sculpture-from-STL folder
- Or open the built-in terminal (View > Terminal) and run:

```
git clone https://github.com/SahanaRajiv/Web-App-Lego-Sculpture-from-STL.git
```

3

### Let Cursor Index Your Project

- Cursor automatically reads and indexes all files when you open the folder
- Takes a few seconds
- After indexing, the AI understands every file, function, and variable

4

### Open the AI Chat

- Press Cmd+L (Mac) or Ctrl+L (Windows)
- Opens the AI chat panel on the right side
- Use it to ask questions, give instructions, or paste error messages

5

### Start Asking Questions

- Try: "Explain what lego\_1.py does"
- Try: "How does the voxelization work in this project?"
- Cursor reads the actual code and explains based on what it sees

## How I Used Cursor to Build This Project

Developing a converter that turns 3D models into LEGO structures was a big undertaking. Cursor was involved in every phase:

### Getting Started

- Set up the project environment
- Chose the right Python libraries for 3D parsing
- Created the initial file structure
- I described what I wanted to build, and it suggested the tools

### Adding Core Features

- Developed voxelization algorithms
- Designed the brick placement logic
- Created the instruction generation system
- I described features in plain English and Cursor wrote the code
- Example: "write a function that checks if each brick has support from below"

### Fixing Complex Bugs

- Pasted error messages into the chat for diagnosis
- Traced through logic to find exact problem lines
- Fixed stability check that was incorrectly removing bricks
- Found a bug where the layer-scanning loop was going the wrong direction

### Documenting My Work

- Wrote clear comments for every function
- Created user-facing guides
- Generated accurate docstrings by reading the actual code

### Refining the Codebase

- Combined duplicate logic
- Renamed unclear variables
- Split large functions into smaller ones
- Made the code easier for others to understand

## Example Prompts for Cursor

You can use Cursor to modify the converter even if you are not an experienced programmer. Here are example prompts you can type into the chat:

### Changing the Model Resolution

*"I want the LEGO model to be bigger and more detailed. Can you increase the voxel resolution from 12 to 20?"*

### Limiting Brick Types

*"I only have 1×1, 1×2, and 1×4 LEGO bricks at home. Can you change the code so it only uses these three sizes?"*

### Customizing Colors

*"Change the instruction colors so that 1×1 bricks are red, 1×2 are blue, 1×4 are green, and 1×6 are yellow."*

### Understanding the Code

*"Explain the voxelize() function line by line. How does it decide which cells are inside the 3D model?"*

### Debugging an Error

*"I'm getting this error: IndexError: index 15 is out of bounds for axis 0 with size 12. What does this mean?"*



# Example Build

## LEGO Rubber Duck

For Kids

Workshop Ready

Printable

This chapter recreates the complete Rubber Duck building instructions. Leaders can print and use it directly during workshops.

To demonstrate how the STL-to-LEGO converter works, I created a Rubber Duck model. Below you'll find everything needed to build it: the input and output files, the complete brick inventory, and step-by-step layer instructions exactly as the converter generated them.

## Input & Output Files

The converter takes a 3D model file (STL format) as input and produces two outputs: a brick-model STL showing what the LEGO version looks like, and a PDF with building instructions.



Input

**Rubber\_Duck.stl**

Original 3D model file



Converter



Output

**Rubber\_Duck\_brick\_model.stl**

LEGO brick version of the model

## Total Bricks Required

Before you start building, gather all the bricks listed below. You need **54 bricks total** in 6 different types.

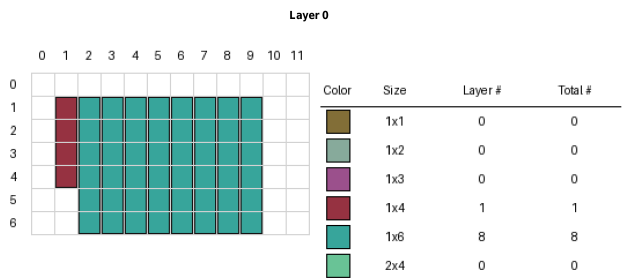
| Color                                                                                               | Size | Total Bricks |
|-----------------------------------------------------------------------------------------------------|------|--------------|
|  Dark Tan        | 1x1  | 4            |
|  Sand Green      | 1x2  | 3            |
|  Medium Lavender | 1x3  | 2            |
|  Dark Red        | 1x4  | 4            |
|  Dark Turquoise  | 1x6  | 39           |
|  Medium Green    | 2x4  | 2            |
| <b>Total</b>                                                                                        |      | <b>54</b>    |

# Layer-by-Layer Building Instructions

The rubber duck model consists of **10 layers** (Layer 0 through Layer 9), built from the bottom up. Each layer diagram shows a 12-column by 7-row grid. The table beside each diagram lists the brick count for that layer and the running total.

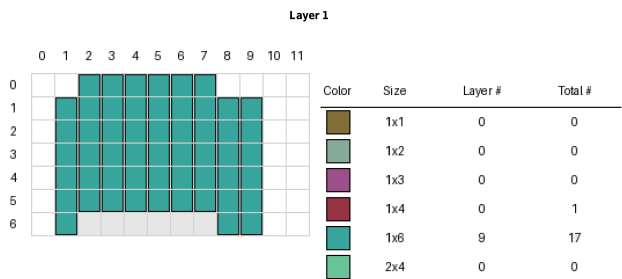
Start with Layer 0 (the base) and stack each subsequent layer on top.

## Layer 0



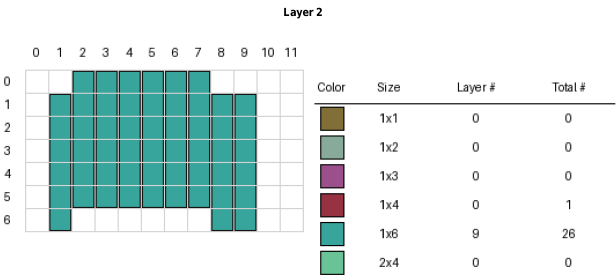
| Color           | Size | Layer | Total |
|-----------------|------|-------|-------|
| Dark Tan        | 1x1  | 0     | 0     |
| Sand Green      | 1x2  | 0     | 0     |
| Medium Lavender | 1x3  | 0     | 0     |
| Dark Red        | 1x4  | 1     | 1     |
| Dark Turquoise  | 1x6  | 8     | 8     |
| Medium Green    | 2x4  | 0     | 0     |
| Subtotal        |      | 9     | 9     |

## Layer 1



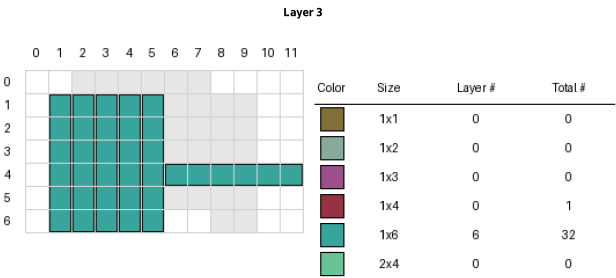
| Color           | Size | Layer | Total |
|-----------------|------|-------|-------|
| Dark Tan        | 1x1  | 0     | 0     |
| Sand Green      | 1x2  | 0     | 0     |
| Medium Lavender | 1x3  | 0     | 0     |
| Dark Red        | 1x4  | 0     | 1     |
| Dark Turquoise  | 1x6  | 9     | 17    |
| Medium Green    | 2x4  | 0     | 0     |
| Subtotal        |      | 9     | 18    |

Layer 2



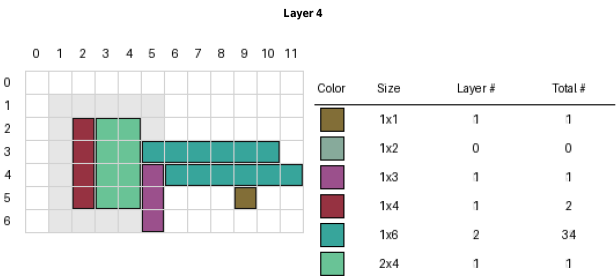
| Color           | Size | Layer | Total |
|-----------------|------|-------|-------|
| Dark Tan        | 1x1  | 0     | 0     |
| Sand Green      | 1x2  | 0     | 0     |
| Medium Lavender | 1x3  | 0     | 0     |
| Dark Red        | 1x4  | 0     | 1     |
| Dark Turquoise  | 1x6  | 9     | 26    |
| Medium Green    | 2x4  | 0     | 0     |
| Subtotal        |      | 9     | 27    |

Layer 3



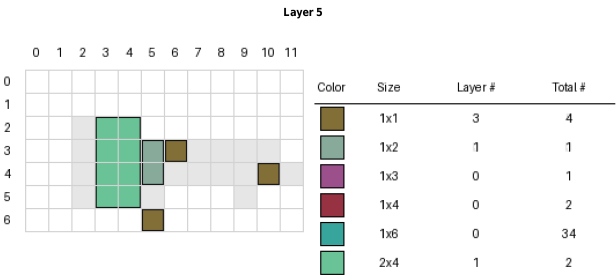
| Color           | Size | Layer | Total |
|-----------------|------|-------|-------|
| Dark Tan        | 1x1  | 0     | 0     |
| Sand Green      | 1x2  | 0     | 0     |
| Medium Lavender | 1x3  | 0     | 0     |
| Dark Red        | 1x4  | 0     | 1     |
| Dark Turquoise  | 1x6  | 6     | 32    |
| Medium Green    | 2x4  | 0     | 0     |
| Subtotal        |      | 6     | 33    |

Layer 4



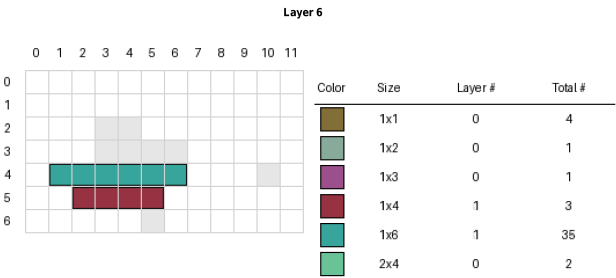
| Color           | Size | Layer | Total |
|-----------------|------|-------|-------|
| Dark Tan        | 1x1  | 1     | 1     |
| Sand Green      | 1x2  | 0     | 0     |
| Medium Lavender | 1x3  | 1     | 1     |
| Dark Red        | 1x4  | 1     | 2     |
| Dark Turquoise  | 1x6  | 2     | 34    |
| Medium Green    | 2x4  | 1     | 1     |
| Subtotal        |      | 6     | 39    |

Layer 5



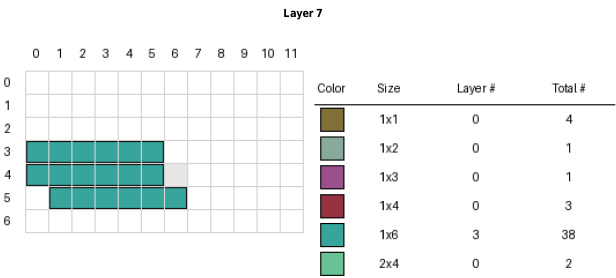
| Color           | Size | Layer | Total |
|-----------------|------|-------|-------|
| Dark Tan        | 1x1  | 3     | 4     |
| Sand Green      | 1x2  | 1     | 1     |
| Medium Lavender | 1x3  | 0     | 1     |
| Dark Red        | 1x4  | 0     | 2     |
| Dark Turquoise  | 1x6  | 0     | 34    |
| Medium Green    | 2x4  | 1     | 2     |
| Subtotal        |      | 5     | 44    |

Layer 6



| Color           | Size | Layer | Total |
|-----------------|------|-------|-------|
| Dark Tan        | 1x1  | 0     | 4     |
| Sand Green      | 1x2  | 0     | 1     |
| Medium Lavender | 1x3  | 0     | 1     |
| Dark Red        | 1x4  | 1     | 3     |
| Dark Turquoise  | 1x6  | 1     | 35    |
| Medium Green    | 2x4  | 0     | 2     |
| Subtotal        |      | 2     | 46    |

Layer 7



| Color           | Size | Layer | Total |
|-----------------|------|-------|-------|
| Dark Tan        | 1x1  | 0     | 4     |
| Sand Green      | 1x2  | 0     | 1     |
| Medium Lavender | 1x3  | 0     | 1     |
| Dark Red        | 1x4  | 0     | 3     |
| Dark Turquoise  | 1x6  | 3     | 38    |
| Medium Green    | 2x4  | 0     | 2     |
| Subtotal        |      | 3     | 49    |

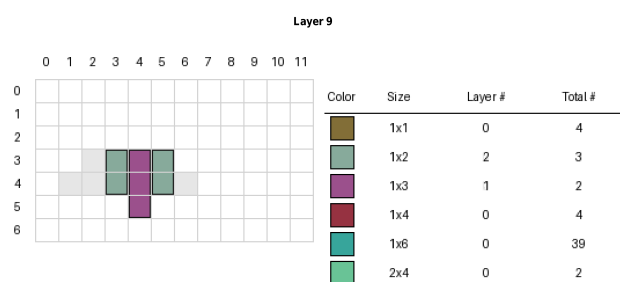


## Layer 8



| Color           | Size | Layer | Total |
|-----------------|------|-------|-------|
| Dark Tan        | 1x1  | 0     | 4     |
| Sand Green      | 1x2  | 0     | 1     |
| Medium Lavender | 1x3  | 0     | 1     |
| Dark Red        | 1x4  | 1     | 4     |
| Dark Turquoise  | 1x6  | 1     | 39    |
| Medium Green    | 2x4  | 0     | 2     |
| Subtotal        |      | 2     | 51    |

## Layer 9



| Color           | Size | Layer | Total |
|-----------------|------|-------|-------|
| Dark Tan        | 1x1  | 0     | 4     |
| Sand Green      | 1x2  | 2     | 3     |
| Medium Lavender | 1x3  | 1     | 2     |
| Dark Red        | 1x4  | 0     | 4     |
| Dark Turquoise  | 1x6  | 0     | 39    |
| Medium Green    | 2x4  | 0     | 2     |
| Subtotal        |      | 3     | 54    |

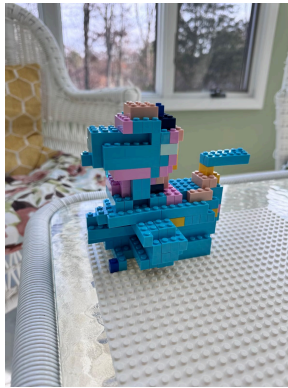
## How to Build

1. **Gather your bricks** -- use the Total Bricks table above to collect all 54 pieces before you start
2. **Start from Layer 0** -- this is the bottom of the duck (the base), built on a flat surface
3. **Follow the grid diagram** -- each colored cell shows where a brick goes; empty cells have no brick
4. **Check the brick table** -- each layer's table shows exactly how many of each brick type you need for that layer
5. **Stack layers in order** -- place each new layer directly on top of the previous one
6. **Press firmly** -- make sure each brick clicks into place before moving to the next layer

**Workshop Tip:** Print each layer diagram on its own page so builders can check off layers as they complete them. Younger builders may benefit from having the bricks for each layer pre-sorted into bags.

## The Finished Model

After completing all 10 layers, your LEGO Rubber Duck should look like the photo below. This example shows how a digital 3D model can be converted into a real, buildable LEGO sculpture using the converter.



The completed LEGO Rubber Duck model

# Host a Workshop

## LEGO STEM Workshop Guide

For Leaders

For Girl Scouts

This chapter is a complete guide for running a STEM workshop using the LEGO converter. Whether you are a troop leader, a Girl Scout earning leadership hours, or a teacher, everything you need is here.

### 1. Materials Needed



#### LEGO Bricks

- Sorted by size: 1x1, 1x2, 1x3, 1x4, 1x6, 2x4
- About 54 bricks per duck model
- Any colors work -- the instructions show sizes, not specific colors
- Extra bricks in case of mistakes or lost pieces



#### Printed Instructions

- One instruction booklet per participant or pair
- Print in color for easiest reading
- Black-and-white works too -- brick sizes are labeled
- Laminate pages if reusing for multiple workshops



#### Computer or Tablet

- Optional -- for demonstrating the converter tool
- One device for the presenter is enough
- Projector or large screen helps for group demos



#### Example STL Files

- Rubber Duck STL included with this project
- Download free models from Thingiverse or MyMiniFactory
- Keep a few options ready for demos



#### Supplies & Setup

- Flat tables for building
- Small bags or cups to sort bricks per participant
- Name tags for larger groups
- Snacks and water (optional but appreciated)



#### Documentation

- Camera or phone for photos
- Feedback forms (paper or digital)
- Sign-in sheet for attendance tracking
- Permission slips if photographing minors



Figure 5.1 -- A completed LEGO model built from the converter, ready for a workshop

## 2. Preparation Checklist

Complete these steps before the workshop day:

### One Week Before

- ✓ Confirm the number of participants
- ✓ Print instruction booklets (one per person or pair)
- ✓ Count and sort bricks -- make sure you have enough for every builder
- ✓ Test the converter tool on your computer to make sure it runs
- ✓ Prepare a short intro script or talking points
- ✓ Set up a sign-up form if tracking attendance

### Day Of

- ✓ Arrive 15–20 minutes early to set up tables and chairs
- ✓ Place a bag of sorted bricks and an instruction booklet at each seat
- ✓ Set up the computer and projector if doing a demo
- ✓ Have the converter tool open and ready (or a pre-recorded demo)
- ✓ Put out a sign-in sheet and name tags
- ✓ Test that the projector or screen is working



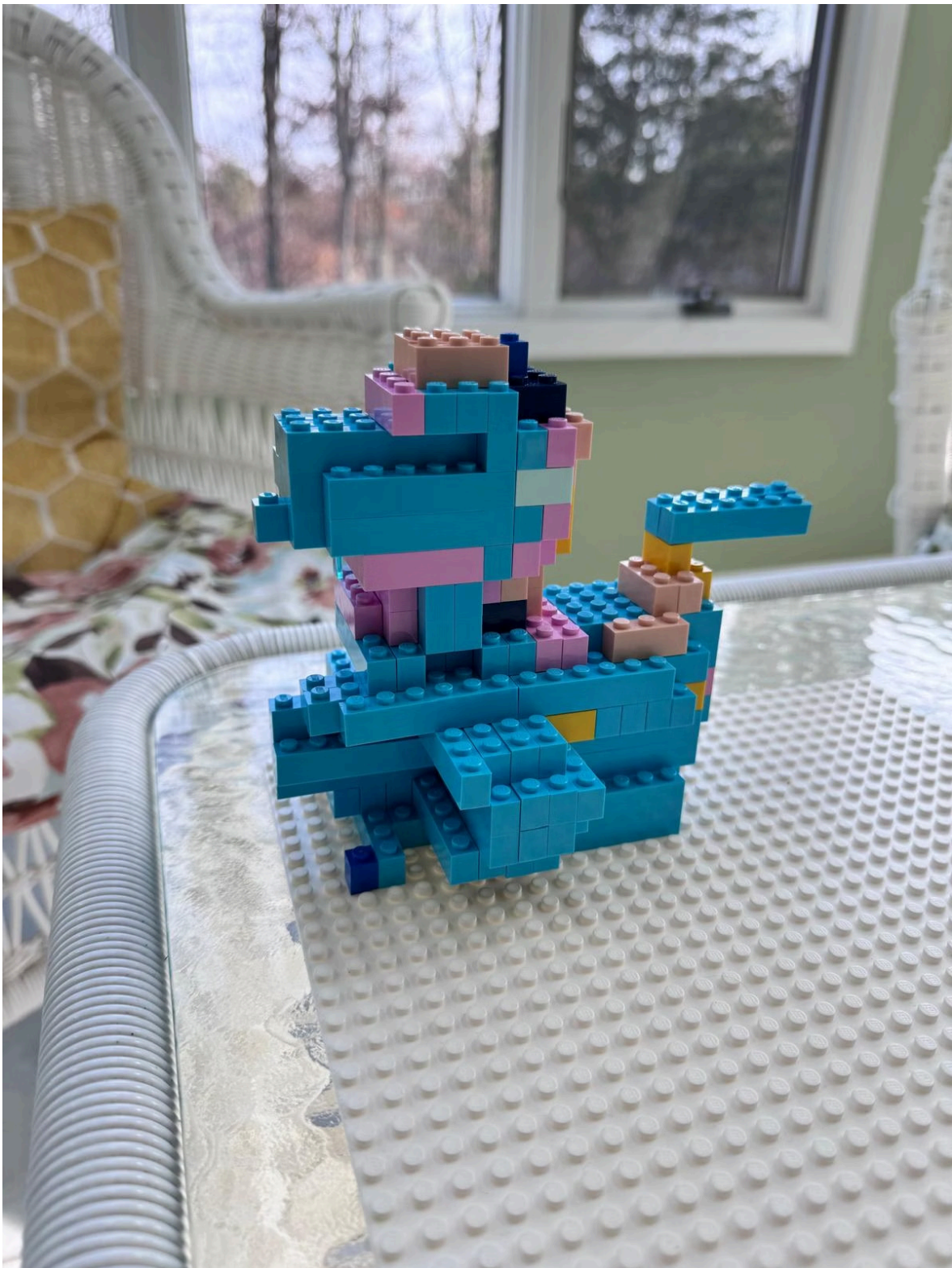


Figure 5.2 -- A completed LEGO Rubber Duck built using the layer-by-layer instructions

### 3. Workshop Schedule

Here is a tested schedule that works well for a one-hour workshop. Adjust times based on your group's age and size.



### Welcome & Introduction 10 min

- Introduce yourself and explain the project
- Ask: "Has anyone used a 3D printer before?"
- Explain: STL files are 3D model files used for 3D printing
- Show the connection: 3D models can become LEGO builds too
- Hand out instruction booklets



### Live Demo 5 min

- Show the converter tool on screen
- Upload the Rubber Duck STL file
- Walk through the output: grid diagrams, brick tables, PDF
- Point out how each layer page matches a real step in the build
- If no computer is available, show the printed instructions and explain each part



### Hands-on Building 30-45 min

- Participants follow the instruction booklet layer by layer
- Start with Layer 0 (the bottom) and work up
- Younger groups: build in pairs or small teams
- Remind them to check the brick inventory table before each layer
- Let them work at their own pace -- some will finish faster



### Show & Tell 10 min

- Ask: "What was the hardest layer?" or "What surprised you?"
- Hand out feedback forms (see section below)



Figure 5.3 -- A completed LEGO duck built during a workshop

## 4. Adapting for Different Ages

### Ages 7–9 (Elementary)

- Use a simpler model with fewer layers (5 or less)
- Pre-sort bricks into individual bags for each builder
- Build in pairs or small teams
- Skip the code demo -- focus entirely on the building
- Allow 45–60 minutes for building time
- Have adult helpers circulate to assist

### Ages 10–13 (Middle School)

- The full Rubber Duck build works perfectly (10 layers, 54 bricks)
- Can build individually
- Include the converter demo -- they will be curious
- Ask them to predict how many bricks they will need before checking the table
- 30–45 minutes is usually enough building time

### Ages 14+ (High School)

- Build the model, then explore the code on a computer
- Show them how to modify the converter (see Chapter 3)
- Challenge them to convert a different STL file
- Introduce Cursor AI as a tool for code exploration
- Can run a longer 90-minute session with code time included

## 5. Collecting Feedback

Hand out a short feedback form at the end of the workshop with 3–5 simple questions. Take photos of the setup and finished models for documentation.



# What I Learned

## Reflections & Future

Through this project, I learned advanced coding skills, problem-solving strategies, and how to teach STEM concepts to others. Here are the key skills and experiences I gained along the way.

## Skills I Developed



### Coding Using AI

Using AI-powered tools like Cursor to write, debug, and understand code



### Documentation

Writing clear guides for users of all technical levels



### Public Speaking

Presenting the project to workshops and community groups



### Problem Solving

Breaking down big challenges into manageable steps

## Challenges I Overcame

One of the biggest technical challenges was solving the **ERR\_SSL\_PROTOCOL\_ERROR** when deploying the app online. My browser required HTTPS (a secure connection), but my app was using HTTP. I learned about SSL certificates and set up a reverse proxy to make the app work securely for everyone.

I also tackled challenges with:

- Making the voxelization algorithm efficient for larger models
- Ensuring LEGO structures are physically stable and buildable
- Generating clear, readable instruction images
- Deploying the web application for public access

# Appendix

## Glossary & Resources

Reference materials, definitions, and links to help you explore the project further.

### Glossary

| Term             | Definition                                                                                      |
|------------------|-------------------------------------------------------------------------------------------------|
| STL              | A 3D model file format that describes the surface of a 3D object using tiny triangles.          |
| Voxel            | A 3D pixel; a small cube representing one unit of space in a 3D model.                          |
| Voxelization     | Converting a smooth 3D model into a grid of small cubes that can be represented by LEGO bricks. |
| Layer            | A horizontal slice of a LEGO build. Models are built layer by layer from bottom to top.         |
| Stability Check  | Ensures every brick is properly supported by bricks below it, preventing floating pieces.       |
| Flask            | A Python web framework used to build the web interface for the converter tool.                  |
| Brick Inventory  | A complete list of every LEGO brick needed to build the model, organized by size and color.     |
| Shell Extraction | A technique to make a model hollow, using bricks only for the outer surface.                    |

### Resources



#### GitHub Repository

Full source code for the STL-to-LEGO converter

<https://github.com/SahanaRajiv/Web-App-Lego-Sculpture-from-STL>



#### Deployed Web App

Live version of the converter tool

<https://web-app-lego-sculpture-from-stl--sahanarajiv.replit.app/>





**My Website**

<https://www.sahanarajiv.com/>



**YouTube Channel**

<https://www.youtube.com/@SahanaRajiv>